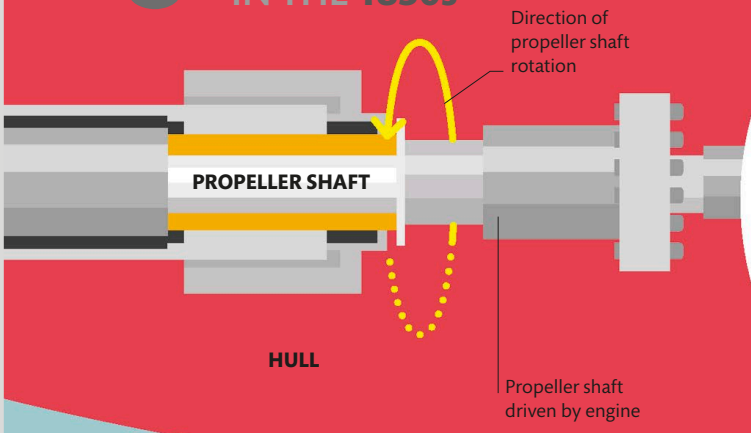
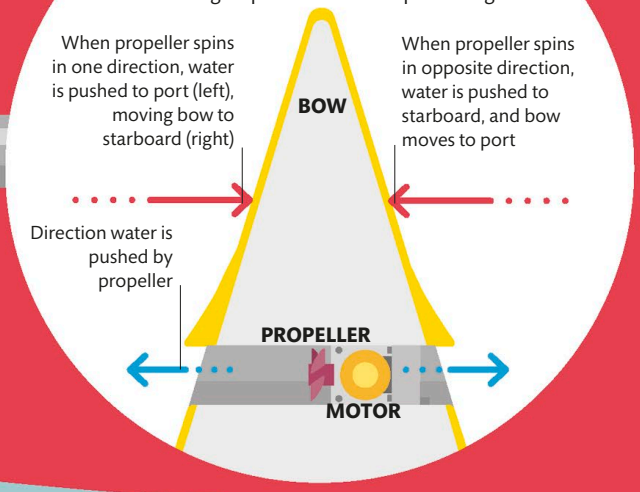
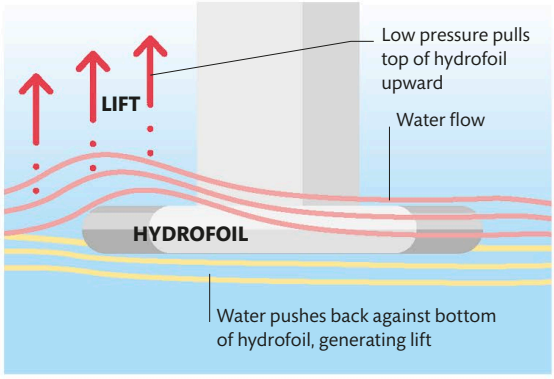
THE FIRST SUCCESSFUL MARINE PROPELLERS WERE DEVELOPED IN THE 1830s



BOW THRUSTER

Some larger vessels have propellers in their bow or stern called thrusters, which are used to create sideways thrust. They enable a vessel to maneuver in tight spaces without help from tugboats.





Hydrofoils

Water pressing against a vessel's hull causes drag, which slows the vessel down since its engine has to work harder to overcome this resistance. Hydrofoil boats minimize drag by using underwater wings called hydrofoils, or foils, which work in the same way as an airplane's wing (see p.62) to lift the whole hull out of the water. Because water is denser than air, compared with an aircraft's wing, a hydrofoil can create more lift at a lower speed.

Types of hydrofoils

Surface-piercing foils break through the surface of the water, while fully submerged foils stay underwater.



SURFACE-PIERCING HYDROFOIL



FULLY SUBMERGED HYDROFOIL

POD PROPULSION

It is common today for large ships to be propelled and steered by devices called azimuth thrusters. These contain an electric motor turning a propeller. The whole pod can be rotated to provide thrust in any direction.

